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BMJ Open

The effectiveness and safety of bee venom pharmacopuncture for rheumatoid arthritis: a systematic review protocol

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Keywords:	RHEUMATOLOGY, COMPLEMENTARY MEDICINE, PAIN MANAGEMENT

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The effectiveness and safety of bee venom pharmacopuncture for rheumatoid arthritis: a systematic review protocol

Won-Suk Sung[1], Jung-Hyun Kim[2], Dong Hyuk Lee[3],
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ABSTRACT

Introduction: Rheumatoid arthritis (RA) is the common autoimmune disease with low quality of life. The representative treatment is medication and medication usage has improved through update of clinical guidelines, however there are still limitations. Bee venom (BV) has been reported to have meaningful therapeutic effects and the possibility of alternative options for RA through several types of studies, but there is no well-organized and recent published systematic review (SR).

Methods: We will search randomized controlled trials about the BV on RA in various databases, manual research, and contacting authors. With screening and reviewing process, we will identify the eligible studies and extract the needed data. The primary outcome will be the disease activity scores indicating the improvement of RA symptoms (American College of Rheumatology response criteria 20, 50, 70), functions (Health Assessment Questionnaire, Disease Activity Score of 28 joints), joint (Western Ontario and McMaster universities osteoarthritis index), pain (Visual Analogue Scale, Numerical Rating Scale), and effective rate. The secondary outcomes will be the RA-related blood test levels and adverse events. We will perform a meta-analysis by Review Manager software, the assessment of risk of bias by Cochrane Collaboration "risk of bias", and the determination of quality of evidence by Grades of Recommendation, Assessment, Development and Evaluation.

Ethics and dissemination: Our SR will suggestthe clinical evidence of the use BV for RA to patient, clinicians, and policymakers. We will publish our results in a peer-review journal.

PROSPERO registration number: CRD42021238058

Keywords: rheumatoid arthritis, bee venom pharmacopuncture, systematic review, meta-analysis

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ARTICLE SUMMARY

Strengths and limitations of this study

- This study will evaluate the effectiveness and safety of bee venom on rheumatoid arthritis overcoming the limitations of existing studies.
- This study will assess more than 10 databases without language limitation.
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INTRODUCTION

Rheumatoid arthritis (RA) is the autoimmune disease characterized by pain, swelling, and joint destruction.[1] In the clinical practice, the criteria for RA has been used for the diagnosis by calculating the score in the section of joints, serology, symptom duration, and acute phase reactants and RA is considered if there are definite swelling joints not explained by another disease, the presence of rheumatoid factor (RF), elevated level of C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR).[2, 3]

In the south Korea, the prevalence of RA increased yearly from 0.28% in 2009 to 0.32% in 2012 with the increasing incidence from 28.5 per 100,000 in 2010 to 42 per 100,000 in 2013.[4] Regarding world-wide, Tayar et al. suggested that RA is the commonest inflammatory polyarthritis with the prevalence of 1% and Safiri et al. reported that the age-standardised prevalence increased by 7.4% and the incidence increased by 8.2% between 1990 and 2017, with the increasing economic loss between 2012 and 2017.[5, 6]

The most recommendable treatment for RA is medication. Since the disease-modifying antirheumatic drugs (DMARDs) were introduced in 1970s,[7] recent clinical guideline has been updated including conventional synthetic, biologic, targeted synthetic DMARDs, and glucocorticoid and provided the recommendable usage in the various clinical settings.[8] However, there are still adverse effects including rash, diarrhea, chronic kidney disease, gastrointestinal distress, and bone marrow suppression.[9]

Bee venom (BV) has been used in various diseases including pain, skin problem, and cancer.[10] It contains various peptides and melittin, a major peptide component, is known to be have anti-inflammatory and anti-arthritis properties.[11] With other pharmaceutical properties including apamin, adolapin, and various enzymes, several studies have reported its effect on RA through experiment, clinical trial, and review.[12-14]

With diverse studies, there are two systematic reviews (SRs) about BV pharmacopuncture on RA, but there are some limitations.[15, 16] One SR included only one randomized controlled trial (RCT) and publication year was old (2014). Another SR was published in 2020 and referred

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several studies but, did not focused on RA. Therefore, the purpose of this SR was to evaluate the effectiveness and safety of BV by including recent published RCTs about RA.

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METHODS

Study design

This SR was designed according to the Preferred Reporting Items for Systemic reviews and Meta-Analyses Protocols (PRISMA-P) 2015 Statement.[17]

Study registration

The protocol was registered in PROSPERO (Registration number: CRD42021238058).

Eligibility criteria

Participants

Patients diagnosed as RA will be included in this SR. In the accordance with the criteria for RA, there is no limitation about age and gender, but we will exclude patients who were diagnosed as other arthritis including osteoarthritis.

Types of interventions

RCTs that used BV as intervention mainly will be included in this SR. The usage of combination treatment with BV in the experimental group should be consistent with control group. Regarding BV, studies comparing the different concentration, treatment duration, dosage, and acupuncture points will be excluded.

Type of comparators

There is no limitation for comparators. Placebo, no treatment, conventional treatments for RA including medication, physical therapy will be eligible.

Type of studies

We will only include RCTs that investigated the effect of BV for RA. Non-RCTs or uncontrolled clinical trials including observational studies, cross-sectional studies, pilot studies, case reports, and SR will be excluded. RCTs that did not provide the randomization method or

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conducted an incorrect randomization will be excluded. There will be no restriction about language or journal.

Outcome measures

Based on the previous SRs about RA,[18-20] disease activity indices that present the degree of RA symptoms will be the primary outcome measure. It includes the indicators about the RA improvement (American College of Rheumatology response criteria (ACR) 20, 50, 70), function (Health Assessment Questionnaire (HAQ), Disease Activity Score of 28 joints (DAS 28)), joint (Western Ontario and McMaster universities osteoarthritis index (WOMAC)), pain (visual analogue scale (VAS), numerical rating scale (NRS)), and effective rate.[21] Blood test about RA including ESR, CRP, RF, anti-cyclic citrullinated peptides, and adverse events will be included as secondary outcome measures.

Information sources and search strategy

We will use the following electronic databases: MEDLINE, EMBASE, Cochrane library, China National Knowledge Infrastructure (CNKI), CiNii, J-STAGE (Japanese database), KoreaMed, Korean Medical Database, Korean Studies Information Service System (KISS), National Digital Science Library (NDSL), Korea Institute of Science and Technology Information (KISTI), and Oriental Medicine Advanced Searching Integrated System (OASIS). Researchers will search the studies from the inception to September 2021 and use search terms about RA (such as rheumatoid arthritis, rheumatoid) and BV (bee venom, bee venom acupuncture, bee venom pharmacopuncture). The search will be performed according to the language provided by each database (Table 1). Regarding additional research, we will conduct manual searching including textbooks and reference lists on retrieved articles. We will also try to contact the corresponding authors if necessary.

Table 1. Search strategy for the Pubmed

No.	Search terms

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#1	rheumatoid arthritis
#2	rheumatoid OR arthritis
#3	#1 and #2
#4	randomized controlled trial OR random* OR placebo
#5	controlled clinical trial OR trial
#6	bee venom
#7	bee venom pharmacopunture OR bee venom acupuncture
#8	#6 OR #7
#9	#3 and (#4 OR #5) and #8

Study selection

Two researchers will perform screening and reviewing for the eligibility. In the screening procedure, studies will be identified by the titles, abstracts, and full-text (if possible). After excluding the duplicates and irrelevant studies, the reviewing procedure for eligibility will be performed by the full-texts (Figure 1). Disagreement will be resolved by the discussion or the mediation by third reviewer.

Figure 1. PRISMA flow diagram

Data management

Studies will be managed by using Endnote X9.

Data extraction

Reviewers will extract the information including first author, publication year, patients' characteristics, intervention in each group (process, treatment site, and period), outcome measures, results, and study quality. Disagreement will be resolved by the discussion or the

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mediation. If necessary, we will contact the corresponding author to obtain the incomplete data. If there is no response, we will describe this omission in the SR.

Data synthesis and analysis

We will use Review Manager software (Version 5.3; Copenhagen; The Nordic Cochrane Center, The Cochrane Collaboration, 2014) for the meta-analysis. The changes from the baseline to the completion of the intervention will be used and the mean difference and 95% confidence intervals (CIs) in same outcome measures, and the standardized mean difference and 95% CI in different outcome measures will be calculated to estimate the effect.

The heterogeneity will be determined by Chi-squared and I-squared tests. The interpretation of the heterogeneity will be as follows: I-squared: 0%–40% unimportant heterogeneity; 30%–60% moderate heterogeneity; 50%–90% substantial heterogeneity; and 75%–100% considerable heterogeneity.

If subgroup analysis is possible, it will be conducted based on the main intervention in control group. If quantitative synthesis is not possible, a narrative synthesis will be conducted using available data. Funnel plot will be applied to assess the publication bias by more than 10 included studies. The Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) method will also be utilized to determine the quality of evidence.[22]

Risk of bias assessment

Reviewers will assess independently the risk of bias by using the Cochrane Collaboration "risk of bias", which consists of seven domains (sequence generation, allocation concealment, blinding of participants and investigators, blinding of outcome assessors, incomplete outcome data, selective outcome reporting, and other biases).[23] Disagreements will be resolved by discussion or medication as indicated above.

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ETHICS AND DISSEMINATION

RA is one of the most common immune disease that results in lower quality of life. Although previous DMARDs have been helpful for RA patients but there are also needs for alternative options. BV had the possibility and several studies have been published. However, existing SR about BV on RA have limitations and this SR will suggest the clinical evidence. Also, we hope our review will be helpful for health policy makers, clinical practitioners, patients, and further research.This review did not require the ethical approval since there was no need for personal information collection and patient recruitment.All of the results will be published in a peer-review journal.

Acknowledgements

Not applicable

Competing Interests

The authors declare that no conflict of interest.

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Author contributions

Joo-Hee Kim is the guarantor of the review. WS Sung and EJ Kim conceived the topic. BK Seo and YH Baek provided the methodology and Jung-Hyun Kim and DH Lee conducted investigation. WS Sung wrote the first draft of the protocol and SU Hong and KH Lee reviewed

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and edited. All authors checked the final version and agreed on the journal to which the article will be submitted.

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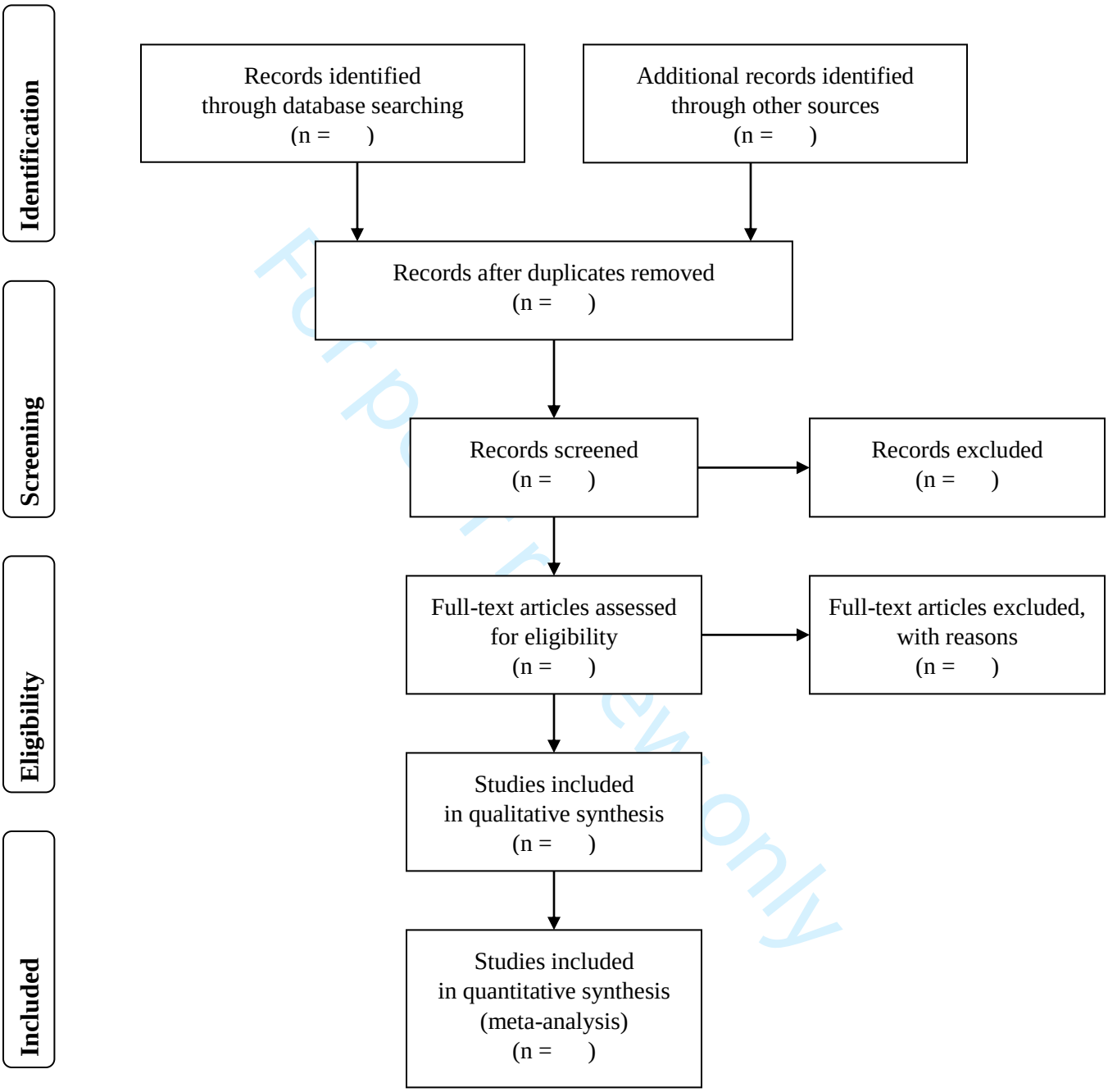
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Figure Legends

Figure 1. PRISMA flow diagram

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2, 6
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	10-11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	10
Sponsor	5b	Provide name for the review funder and/or sponsor	10
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	10
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-5
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8-9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9-

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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PROSPERO registration number: CRD42021238058

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Rheumatoid arthritis (RA) is the autoimmune disease characterized by pain, swelling, and joint destruction.[1] In the clinical practice, the criteria for RA has been used for the diagnosis by calculating the score in the section of joints, serology, symptom duration, and acute phase reactants and RA is considered if there are definite swelling joints not explained by another disease, the presence of rheumatoid factor (RF), elevated level of C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR).[2, 3]

In the south Korea, the prevalence of RA increased yearly from 0.28% in 2009 to 0.32% in 2012 with the increasing incidence from 28.5 per 100,000 in 2010 to 42 per 100,000 in 2013.[4] Regarding world-wide, Tayar et al. suggested that RA is the commonest inflammatory polyarthritis with the prevalence of 1% and Safiri et al. reported that the age-standardised prevalence increased by 7.4% and the incidence increased by 8.2% between 1990 and 2017, with the increasing economic loss between 2012 and 2017.[5, 6]

The most recommendable treatment for RA is medication. Since the disease-modifying antirheumatic drugs (DMARDs) were introduced in 1970s,[7] recent clinical guideline has been updated including conventional synthetic, biologic, targeted synthetic DMARDs, and glucocorticoid and provided the recommendable usage in the various clinical settings.[8] However, there are still adverse effects including rash, diarrhea, chronic kidney disease, gastrointestinal distress, and bone marrow suppression.[9]

Bee venom (BV) has been used in various diseases including pain, skin problem, and cancer.[10] It contains various peptides and melittin, a major peptide component, is known to be have anti-inflammatory and anti-arthritis properties.[11] With other pharmaceutical properties including apamin, adolapin, and various enzymes, several studies have reported its effect on RA through experiment, clinical trial, and review.[12-14]

With diverse studies, there are two systematic reviews (SRs) about BV pharmacopuncture on RA, but there are some limitations.[15, 16] One SR included only one randomized controlled trial (RCT) and publication year was old (2014). Another SR was published in 2020 and referred

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several studies but, did not focused on RA. Therefore, the purpose of this SR was to evaluate the effectiveness and safety of BV by including recent published RCTs about RA.

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METHODS

Study design

This SR was designed according to the Preferred Reporting Items for Systemic reviews and Meta-Analyses Protocols (PRISMA-P) 2015 Statement.[17]

Study registration

The protocol was registered in PROSPERO (Registration number: CRD42021238058).

Eligibility criteria

Participants

Patients diagnosed as RA will be included in this SR. In the accordance with the criteria for RA, there is no limitation about age and gender, but we will exclude patients who were diagnosed as other arthritis including osteoarthritis.

Types of interventions

RCTs that used BV as intervention mainly will be included in this SR. The usage of combination treatment with BV in the experimental group should be consistent with control group. Regarding BV, studies comparing the different concentration, treatment duration, dosage, and acupuncture points will be excluded.

Type of comparators

There is no limitation for comparators. Placebo, no treatment, conventional treatments for RA including medication, physical therapy will be eligible.

Type of studies

We will only include RCTs that investigated the effect of BV for RA. Non-RCTs or uncontrolled clinical trials including observational studies, cross-sectional studies, pilot studies, case reports, and SR will be excluded. RCTs that did not provide the randomization method or

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conducted an incorrect randomization will be excluded. There will be no restriction about language or journal.

Outcome measures

Based on the previous SRs about RA,[18-20] disease activity indices that present the degree of RA symptoms will be the primary outcome measure. It includes the indicators about the RA improvement (American College of Rheumatology response criteria (ACR) 20, 50, 70), function (Health Assessment Questionnaire (HAQ), Disease Activity Score of 28 joints (DAS 28)), joint (Western Ontario and McMaster universities osteoarthritis index (WOMAC)), pain (visual analogue scale (VAS), numerical rating scale (NRS)), and effective rate.[21] Blood test about RA including ESR, CRP, RF, anti-cyclic citrullinated peptides, and adverse events will be included as secondary outcome measures.

Information sources and search strategy

We will use the following electronic databases: MEDLINE, EMBASE, Cochrane library, China National Knowledge Infrastructure (CNKI), CiNii, J-STAGE (Japanese database), KoreaMed, Korean Medical Database, Korean Studies Information Service System (KISS), National Digital Science Library (NDSL), Korea Institute of Science and Technology Information (KISTI), and Oriental Medicine Advanced Searching Integrated System (OASIS). Researchers will search the studies from the inception to 31 May 2022 and use search terms about RA (such as rheumatoid arthritis, rheumatoid) and BV (bee venom, bee venom acupuncture, bee venom pharmacopuncture). The search will be performed according to the language provided by each database (Table 1). Regarding additional research, we will conduct manual searching including textbooks and reference lists on retrieved articles. We will also try to contact the corresponding authors if necessary.

Table 1. Search strategy for the Pubmed

No.	Search terms

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#1	rheumatoid arthritis
#2	rheumatoid OR arthritis
#3	#1 and #2
#4	randomized controlled trial OR random* OR placebo
#5	controlled clinical trial OR trial
#6	bee venom
#7	bee venom pharmacopuncture OR bee venom acupuncture
#8	#6 OR #7
#9	#3 and (#4 OR #5) and #8

Study selection

Two researchers will perform screening and reviewing for the eligibility. In the screening procedure, studies will be identified by the titles, abstracts, and full-text (if possible). After excluding the duplicates and irrelevant studies, the reviewing procedure for eligibility will be performed by the full-texts (Figure 1). Disagreement will be resolved by the discussion or the mediation by third reviewer.

Figure 1. PRISMA flow diagram

Data management

Studies will be managed by using Endnote X9.

Data extraction

Reviewers will extract the information including first author, publication year, patients' characteristics, intervention in each group (process, treatment site, and period), outcome measures, results, and study quality. Disagreement will be resolved by the discussion or the

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mediation. If necessary, we will contact the corresponding author to obtain the incomplete data. If there is no response, we will describe this omission in the SR.

Data synthesis and analysis

We will use Review Manager software (Version 5.3; Copenhagen; The Nordic Cochrane Center, The Cochrane Collaboration, 2014) for the meta-analysis. The changes from the baseline to the completion of the intervention will be used and the mean difference and 95% confidence intervals (CIs) in same outcome measures, and the standardized mean difference and 95% CI in different outcome measures will be calculated to estimate the effect. In the data analysis, a random-effects model will be based because the included studies may come from different population.

The heterogeneity will be determined by Chi-squared and I-squared tests. The interpretation of the heterogeneity will be as follows: I-squared: 0%–40% unimportant heterogeneity; 30%–60% moderate heterogeneity; 50%–90% substantial heterogeneity; and 75%–100% considerable heterogeneity.

If subgroup analysis is possible, it will be conducted based on the main intervention in control group. If quantitative synthesis is not possible, a narrative synthesis will be conducted using available data. Funnel plot will be applied to assess the publication bias by more than 10 included studies. The Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) method will also be utilized to determine the quality of evidence.[22]

Risk of bias assessment

Reviewers will assess independently the risk of bias by using the Cochrane Collaboration "risk of bias", which consists of seven domains (sequence generation, allocation concealment, blinding of participants and investigators, blinding of outcome assessors, incomplete outcome data, selective outcome reporting, and other biases).[23] Disagreements will be resolved by discussion or medication as indicated above.

Patient and public involvement

No patient involved

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ETHICS AND DISSEMINATION

RA is one of the most common immune disease that results in lower quality of life. Although previous DMARDs have been helpful for RA patients but there are also needs for alternative options. BV had the possibility and several studies have been published. However, existing SR about BV on RA have limitations and this SR will suggest the clinical evidence. Also, we hope our review will be helpful for health policy makers, clinical practitioners, patients, and further research. This review did not require the ethical approval since there was no need for personal information collection and patient recruitment. All of the results will be published in a peer-review journal.

Acknowledgements

Not applicable

Competing Interests

The authors declare that no conflict of interest.

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Author contributions

Joo-Hee Kim is the guarantor of the review. WS Sung and EJ Kim conceived the topic. BK Seo and YH Baek provided the methodology and Jung-Hyun Kim and DH Lee conducted investigation. WS Sung wrote the first draft of the protocol and SU Hong and KH Lee reviewed

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and edited. All authors checked the final version and agreed on the journal to which the article will be submitted.

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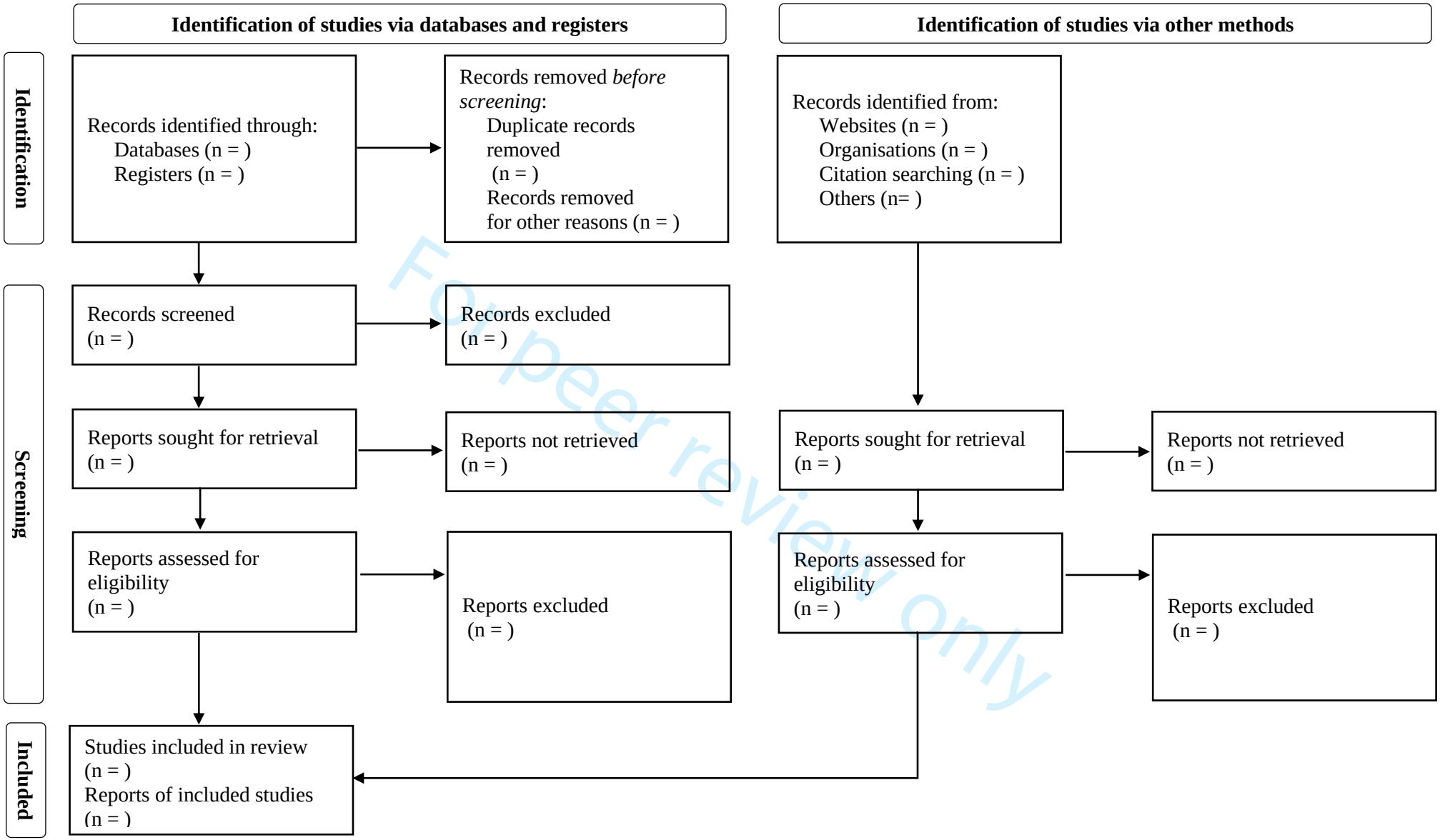
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Figure Legends

Figure 1. PRISMA flow diagram

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2, 6
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	10-11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	10
Sponsor	5b	Provide name for the review funder and/or sponsor	10
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	10
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-5
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8-9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9-

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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